

Module 8

I²C — UVM+SV Verification

Learning objectives

- I²C UVM agent: I2cTransaction, I2cSequence, I2cDriver, I2cMonitor, I2cScoreboard.
- Interface: i2c_if (clk, rst_n, start, addr, data_in, scl, sda, clk_div_tick, busy, done) connects U...
- How to hook up a UVM monitor and scoreboard to a bus-oriented DUT.

Prerequisites

- Module 7 (I²C protocol + RTL + basic TB).
- Verilator, Make, C++ compiler, UVM (UVM_HOME or vendored UVM).

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

flowchart LR

T1["I²C UVM Agent"]

T2["Toolchain"]

T1 --> T2

T2 --> N["Next module in course"]

Extend I²C verification to ****UVM+SV**** — I²C agent (transaction, sequence, driver, monitor, scoreboard); run on Verilator.

Overview

- Module 8 builds on Module 7 (I²C protocol + RTL + basic testbench):

Syllabus topics

1. I²C UVM Agent

- Transaction (I2cTransaction): address and data to write; monitor fills observed_addr/observed_data.
- Sequence (I2cSequence): produces directed transactions (e.g. same addr, different data bytes).
- Driver (I2cDriver): gets transaction, drives start/addr/data_in for one cycle, waits for done.
- Monitor (I2cMonitor): watches SCL/SDA; after START, samples SDA on rising SCL to reconstruct 8 bits...
- Scoreboard (I2cScoreboard): compares expected (from test) vs observed_addr/observed_data.

2. Toolchain

- Same as Module 6 (SPI UVM): Verilator with -sv, --timing, --trace, UVM include paths, `make SIM=ver...

Hands-on examples

Module 8 self-check

```
$ ./scripts/module8.sh --check
```

Terminal: module8_check

```
[[0;36m===== [[0m
[[0;36mModule 8: Demo commands (I²C UVM example) [[0m
[[0;36m===== [[0m

From repo root:

# 1) Environment sanity:
verilator --version
make --version
echo "$UVM_HOME"

# 2) Run the I²C UVM example (inside the example directory):
cd module8/examples/i2c_uvm
make SIM=verilator TEST=test_i2c_uvm

# 3) Inspect logs and waveforms:
ls obj_dir
# If a VCD is produced:
ls *.vcd

[[1;33m[INFO] [[0m See module8/EXAMPLES.md and module8/examples/i2c_uvm/README.md for more details.
```

Example 1: I²C UVM

- UVM phases and reporting (DRIVER, MONITOR, SCOREBOARD).
- Scoreboard checks (expected vs observed address+data from the bus).
- Final scoreboard summary (matches and mismatches).

Demo: I²C UVM

```
$ ./scripts/module8.sh --run
```

Terminal: ex01_i2c_uvm

Success

You should see UVM phases, DRIVER/MONITOR/SCOREBOARD messages, and a scoreboard summary with all wri

Practice & assessment

Exercises

- Run i2c_uvm
- Trace one transaction
- Change the sequence
- Compare across protocols

Assessment checklist

- Can describe the I²C UVM agent and how it maps to I²C (addr+data, monitor on SCL/SDA).
- Can run i2c_uvm and interpret UVM output.
- Can compare UVM structure across UART, SPI, and I²C.

Summary & next steps

- Extend I²C verification to ****UVM+SV**** — I²C agent
(transaction, sequence, driver, monitor, scoreboa...
- Complete module8/CHECKLIST.md
- Run `./scripts/module8.sh --check`
- Next: Next module in course